Department of Electronics & Telecommunication Engineering

Cyber Security (Batch 2023-27, Semester VII) - Continuous Assessment II: Case Study (Experiential Learning)

Digital Arrest Fraud Against Senior Citizens: A Case Study of the Pune Digital Arrest Scam (January 2025) and DA-Shield, a Layered Detection and Prevention Framework

Group No.: 31

PRN	Student Name
23070123002	Aamir Ayaan Quraishi
23070123026	Arav Upadhyay
23070123122	Shivendra Dwivedi
23070123164	Pranjal Panwar

Topic Selected: Pune Digital Arrest Scam (January 2025) targeting a 78-year-old woman, resulting in Rs. 4.37 crore loss through WhatsApp and phone impersonation.

Faculty: Dr. Mangal Singh

Abstract

A 78 year old woman of Pune in January of 2025 lost Rs. They called her on the phone and WhatsApp, claiming to be from the police and the central investigation department, and convinced her that she was being “arrested” via digital means and that she should transfer her savings to their account.4.37 crore after fraudsters posing as police or CI officers phones her and WhatsApped her that she was being “arrested” digitally and that her savings should be transferred to their account. Although "digital arrest" is not a legal concept in India, reported cases of this scam increased from 39,925 cases in 2022 to 1,23,672 cases in 2024 with the reported losses increasing from Rs. 91.14 crore to Rs. 1,935.51 crore [1]. This report has been structured as a six-stage social engineering kill chain, maps each stage to the persuasion principle it leverages, as well as to the telecom, messaging, banking, incident response and awareness technical or institutional control that failed to stop it, and proposes DA-Shield as a five-layer defence framework that runs across telecom, messaging, banking, incident response and awareness. In Python two components were implemented and evaluated. For digital-arrest scripts at the script level, Module A achieved an F1 score of 0.84 and a recall of 0.95 for a synthetic 910-message corpus, as opposed to 0.66 for a baseline built around a keyword rule. In module B, an explainable risk engine for senior citizens' high-value transfers, 95.0% of the fraudulent instalments were caught with a false-positive rate of 1.07% on 22,605 simulated transactions, and 122 of 150 simulated campaigns were thwarted at the first instalment. The engine would have deposited all of the instalments including the first one on a 24-hour hold with a trusted-contact callback mode. The findings suggest the final control point is the bank counter and payment rail where the victim's isolation can be reduced prior to the departure of the money.

Keywords: *Digital arrest; social engineering; vishing; impersonation fraud; elder financial abuse; WhatsApp fraud; mule accounts; transaction risk scoring; scam text classification; cyber security policy*

1. Introduction

India's digital payments revolution has now given its citizens quick and cheap access to their funds – but it has also reduced the time a fraudster has to access them. The National Cyber Crime Reporting Portal has received a number of complaints about the "digital arrest" scam, which has emerged at a rapid pace among the many cyber-enabled frauds. Criminals call the victims with a continuous audio or video call, claiming to be of the police, the Central Bureau of Investigation (CBI), the Enforcement Directorate (ED), customs or the telecom regulator, and accuse them of committing a serious crime and keep them on a continuous audio and video call until an investigation is over. The victim will be told to not say anything about the incident and to "transfer money for the security deposit" or to "clear their name". The police have emphasised that no such practice prevails, for instance, there are no police officers who can be arrested over phone or video calls or request money transfers to resolve cases [2] [3]. The sheer scale of the issue is now reflected in official statistics. The Ministry of Home Affairs responded in writing to the Rajya Sabha on 12 March 2025 that it had received reports of 39,925 numbers of "digital arrest" incidents during 2022 and 60,676 in 2023 and 1,23,672 in 2024, showing that the number of reported incidents defrauded rose during the year. 91.14 crore to Rs. 339.03 crore and then to Rs. 1,935.51 crore [1]. Incidents doubled approximately every 12 months over two years and losses increased by nearly twenty-one times as well [4] – this means that more money is being wiped out with each successful attack. This is illustrated in Fig. 1. A total of 17,718 cases worth Rs. were reported during the first two months of 2025 alone. 210.21 crore were reported [1], [5].

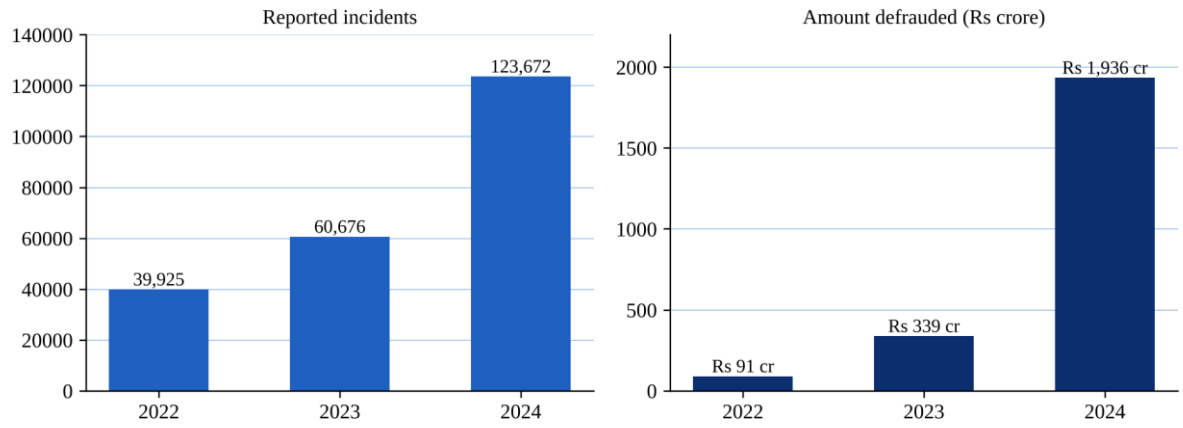


Fig. 1. Digital arrest and related incidents reported on the NCRP and the amount defrauded, 2022-2024. Data: Ministry of Home Affairs reply in the Rajya Sabha, 12 March 2025 [1].

Pune is one of the most blatant illustrations of the impact of this fraud on the elderly. As per the data revealed by Pune Cyber Police in September 2026, net loss to cybercriminals on digital platforms with respect to the senior citizens has been pegged at over Rs. 136 from January 2025 to July 2026. One hundred crores; Rs. 85 of these cases. 53.52 crore, were registered in 2025 [3], [6]. In other incidences that happened in Pune, see how big and how damaging individual losses can be: An 82-year old man lost Rs. from his home address Bhandarkar Road. The man, 10.74 crore, and his wife were forced to participate in video calls for hours and suffered a loss of approximately Rs. for about a month before he suffered a heart attack [7], [8], [6] and passed away a few days after that heart attack. 1.2 crore [9].

1.1 The Case Under Study

The case assigned to Group 31 is the Arrest scam in which an 78 year old lady had lost Rs. 4.37 crore after receiving calls from fraudsters passing off themselves as officials, on normal mobiles and WhatsApp. In comparison, the average loss in the 85 senior citizen digital arrest cases registered in Pune during 2025 was approximate Rs. This one is about 7 times of the average of the City for the year (63 lakh [3]). Compared to the average in the country which is around Rs. It is around 280 times higher

at almost 1.57 lakh per incident in 2024 [1]. The case is the high end of the target: an older person with significant savings, who is largely self-sufficient, and whose daily life may be the subject of an organised attack by someone willing to invest days of effort in one person. The authors were unable to find any detailed reports on the public that were made of this person complaint i.e. police station, instalment programmes and arrests. Therefore, the facts provided in the problem statement serve as the given points of the case (location, month, age of victim, amount lost, the modus operandi of the phone and WhatsApp impersonation) and the remaining elements of the attack were constructed based on the case modus operandi documented in similar cases in Pune and Maharashtra during the same period as in this case, [2], [9], [7], [3]. If a detail is reconstructed (not reported), it is marked.

1.2 Problem Statement

A phone and a messaging application were enough to trick a senior citizen, who had no criminal entanglements, into transferring Rs. Over the course of several days, she was handing over 4.37 crore of her money to criminals—and not a single person, neither her family, the police nor the telecom company, the messaging company or the bank, was there to check in on her before she lost the money. This case study therefore has multiple points of failure: The caller was not recognized; the victim was psychologically isolated from others who could debunk the story; and high dollar transfers were made without meaningful friction. The solution needs to cleanse this chain at one or more points such that it is feasible, viable and usable for Indian banks and families and cannot rely on the victim learning about the scam while under pressure.

1.3 Objectives

To create a structured kill chain for the Pune digital arrest attack and document the psychological and technological leverage applied at various stages of the attack. To determine which controls were absent or not effective (telecom, platform, banking, legal and social), through the use of official data and published literature. To create DA-Shield, a multi-layered solution consisting of prevention and detection controls at points where they will make the biggest impact. To build and test two DA-Shield components: a text detector of scam-scripts (Module A); a transaction risk engine for elderly customers (Module B); and to replay the captured case through Module B (Module C). The rest of the report is organised as follows. This Section 2 aims to review the literature related to the social engineering, telephony spoofing, scam victimisation among older adults, fraud detection and the legal response in India. In Section 3, the methodology is presented, which comprises the attack reconstruction and the design of DA-Shield. The results are presented and discussed in Section 5 and conclusions are presented in Section 6.

2. Literature Review

2.1 Social Engineering and the Psychology of Persuasion

Social engineering is the act of exploiting humans instead of machines to release information, access or funds. According to Cialdini's research on influence, there are a few core rules that tend to make people more compliant; scarcity (urgency), commitment and consistency, authority and social proof are just a few. [10] The key elements of the digital arrest scam are authority (the uniformed “officer” and the police-station background), as well as urgency (arrest “within two hours”). Hadnagy outlines how a professional social engineer, when using a believable, invented scenario known as pretexting, combined with elicitation and rapport, can drop defenses on a target [11]. Mouton, Leenen and Venter defined social engineering attacks in a repeatable template of phase (information gathering, relationship development, exploitation and execution), and demonstrated that most attacks in the real world conform to this template [12]. Krombholz et al. categorised the advanced social engineering attacks based on the

channel, the operator and the type of attack, and pointed out that the pretexting attack on the phone is still very successful due to the fact that it might give the victim little time to think [13]. Salahdine and Kaabouch surveyed defence that could be technical and awareness and defence that could be human-centred and found that there was no single defence, but it should be coupled with awareness, detection and process controls [14]. In his empirical study, Workman observed that the individuals who were more committed, more trustful and more accepting of authority had a higher acceptance of pretext attacks [15]. The results of this investigation have informed this report on the need to keep the victim out of the defendant's line of sight, the need to ensure that a frightened person will comply, and the lack of any blame to be placed upon the victim. A number of models have been proposed for intrusion campaigns, including the "kill chain" described by Hutchins, Cloppert and Amin for the security industry [16] who state that an attack is a series of stages, with a defender who acts at, or breaks, any one of the stages defeating the entire attack. This idea is adapted in Section 3 to a social engineering kill chain for digital arrest fraud.

2.2 Scam Victimisation and Older Adults

Stajano and Wilson studied numerous actual scams and concluded that there are seven principles exploited by scammers: distraction, social compliance, herd effect, dishonesty, kindness, need and greed, and time pressure which should all be taken into account rather than attempting to circumvent by asking users to defy the tendencies of human nature [17]. Similarly, Whitty's online romance scam stage model revealed a consistency in how fraudsters intentionally build up the level of commitment over time such that every payment increases the likelihood of the next [18] and this consistency can be seen in the repeated payments and instalments made in digital arrest cases. Button, Lewis and Tapley's review of typologies of frauds was particularly interested in the long-term emotional and financial damage that can occur from fraud and the shame factor to not report fraud [19]. Special interest in older adults. A systematic review and meta-analysis by Burnes et al. reported that in the United States, the prevalence of financial fraud in older adults is approximately 5.4% and is highly variable across studies [20]. James, Boyle and Bennett concluded that for older people not affected by dementia, reduced psychological well-being, reduced social support and reduced literacy were related to increased susceptibility to scams [21]. Spreng, Karlawish and Marson proposed a model that relates changes in cognition, social network and brain function to vulnerability to financial exploitation that increases with age [22], and Ebner et al. reported that the susceptibility to online deception increases as a person gets older and is related to cognition and socio-emotional factors [23]. Combined, these studies provide a ‘psychological’ reason for why the scam is particularly devastating to older victims: it takes away that social support away from victims.

2.3 Caller-ID Spoofing and Telephony Fraud

Tu et al. surveyed the defenses against unwanted and fraudulent calls and found that spoofing the Caller ID is very easy through voice-over-IP gateways and so the number shown on a handset is not a strong indicator of who is calling [24]. In India, calls made from overseas using calling line identification to display local mobile number were reported as activities of fraudsters in various crime activities such as fake digital arrest, courier drug scam, and impersonation of government and police officials by DoT in May 2024, requesting the operators to block such calls [26]. Later, DoT claimed that over $1.35 crore fake incoming calls were blocked during the first day of the launch of its International Incoming Spoofed Calls Prevention System, with 90% of such calls blocked during the first day of the system [26]. These aspects of the measures are important for this case since the initial call in a digital arrest is almost always a local and official call.

2.4 Forged Documents, Video Staging and Deepfakes

Scammers usually set up fake police-station backgrounds, outfits, and fake documents to make it look real when the victim is put on a video call. In his paper towards deepfakes, Westerlund describes how synthetic audio and video have become more affordable and can diminish the trust in what we see and hear [27]. Despite the absence of deepfakes, forged paperwork has been successful; on 17 October 2025, the Supreme Court of India suo motu noted that fraudsters were sending fake Supreme Court Orders with fake signatures and seals on WhatsApp and video calls during the digital arrest scam [28].

2.5 Financial Fraud Detection and Mule Accounts

Abdallah, Maarof, and Zainal classify fraud detection systems into rule-based, statistical and machine learning-based, and urge to highlight the highly imbalanced nature of fraud data and the always present concept drift of criminals adapting [29]. Dal Pozzolo et al. learned that to deploy card-fraud detection in practice, it is important to have precise alerts as investigators can only review a limited number of alerts per day [30]. When the labelled fraud data is limited, Liu, Ting and Zhou's Isolation Forest provides a prevalent unsupervised solution to the problem of detecting anomalous transactions. The funds collected in digital arrest scams are transferred via “mule” account(s): an account opened or rented by someone else and then “passes” the victim's money on. In a Rs. In 2021, Maharashtra Cyber estimated that it had uncovered over 6,500 bank accounts spread over 13 layers through which the stolen funds were laundered in a 58 crore digital arrest case [32]. In the Rs. The police had said they had opened an account in the victim's name and arrested local young men, including a social media influencer, for their involvement in the money trail in a 10.74 crore rupees case [8]. The Reserve Bank of India (RBI) has hit back with MuleHunter.The Reserve Bank Innovation Hub has developed a machine learning model called AI, which banks must use to flag suspected mule accounts [33]; In addition, the 2024 Master Directions on fraud risk management compel banks to implement early-warning systems and report frauds swiftly [34]. However, mule detection works on the receiving account. The attention in this work is given to the sending end where the victim's own bank can block the funds prior to their exit from their system.

2.6 Text-Based Scam Detection

An important application of detecting fraudulent messages is related to SMS spam filtering. Almeida, Hidalgo and Yamakami presented a popular SMS Spam collection and demonstrated that word features are effective in building simple classifiers for SMS spam [35]. Salton and Buckley developed TF-IDF weighting, which is a formidable baseline for representing short texts [36] and can be implemented in standard, well-tested ways in the scikit-learn library as well as via Naive Bayes, logistic regression and random forest [37]. The corpora used in this study, however, are not public scam corpora, but rather corpora developed for a specific task, as these corpora are not dominated by marketing spam are not in the appropriate scripts, namely Hinglish and authoritatives.

2.7 Legal and Institutional Response in India

Digital arrest offences are covered by the Information Technology Act, 2000 (mainly Section 66C: The offence of identity theft and Section 66D: Cheating by personation by means of a computer resource) [38] and Bharatiya Nyaya Sanhita, 2023 (cheating (Section 318), cheating by personation (Section 319), and extortion (Section 308)) [39]. The 1930 helpline, along with the Indian Cyber Crime Coordination Centre (I4C) was responsible for blocking over 3,962 Skype IDs and 83,668 WhatsApp accounts used for digital arrest before early 2025, and had been featured in the Prime Minister's "Mann Ki Baat" speech on 27 October 2024, in which the government spoke about digital arrest [1]. Since then, the

judiciary has taken a more active role, with the Supreme Court ordering a nation-wide investigation into digital arrested scams and asking all states to establish cyber-crime coordination centres, linked with I4C in December, 2025 [40, 41]. The Court in July 2026 had indicated that the Union government should consider differentiating digital arrest from the current regime and impose stiffer penalties for committing digital crimes; it pointed out that WhatsApp's report also indicated that 9,400 digital accounts connected with scams had been banned in just 12 weeks by the platform since January 2026 [42]. In fact, worldwide, the FBI's Internet Crime Complaint Center identifies government impersonation as a specific type of fraud.Moreover, the pattern is not unique to India; the FBI's Internet Crime Complaint Center (IC3) internationally recognizes government impersonation as a category of fraud [43].

2.8 Research Gap

The most salient previous research is summarised in Table I. The literature provides a good account of the reasons why digital arrest scams are successful and what component parts of the ecosystem can be hardened, but there are three gaps. First, most technical solutions have either been designed to deal with the call (spoof blocking) or the receiving account (mule detecting). Few solutions have been developed to deal with the interaction of the sending bank with an old account holder who is now being coached in real time. Secondly, no evaluations have been carried out to test the scam-text classifiers on Indian digital arrest scripts or to test their capability of distinguishing between scam and legitimate police warning scripts composed in the same vocabulary. Thirdly, few studies link these controls together in a framework that correlates each control to a stage of the attack. This report on the three gaps.

Table I. Summary of key related work and the gap addressed by this study

Work	Focus and key finding	Relevance / gap
Cialdini [10]; Workman [15]	Authority and urgency drive compliance; trusting, authority-responsive people are more vulnerable	Explains victim behaviour; offers no technical control
Stajano and Wilson [17]	Seven principles exploited by scammers; design systems around them	Supports placing friction in the payment flow instead of relying on victims
Burnes et al. [20]; James et al. [21]	Fraud is common among older adults; low social support raises risk	Isolation is the core tactic of digital arrest
Tu et al. [24]; DoT [26]	Caller ID is spoofable; India blocks about 90% of spoofed international calls	Fraudsters move to WhatsApp and international numbers
Abdallah et al. [29]; Dal Pozzolo et al. [30]	Rule and ML fraud detection; precision of alerts matters	Not tuned to coerced, authorised payments by seniors
RBI MuleHunter.AI [33]	ML detection of mule (receiving) accounts	Acts after money leaves the victim
Almeida et al. [35]	Classifiers work well on SMS spam	No Indian digital arrest scripts; no hard negatives

3. Methodology

3.1 Research Design

This study uses three techniques. The qualitative case reconstruction sets out how the Pune attack most probably played out. A design-science step then converts the failure points found into a set of controls (DA-Shield). Finally, two of those controls are implemented experimentally and their performance is evaluated on synthetic data based on known scam patterns. Such a combination was selected as the simulated messages will be used as signals to test the solution, and will be enough to understand the attack, but not enough to test a solution in the case itself.

3.2 Data Sources

Official data from the Ministry of Home Affairs (MHA) in reply to Rajya Sabha Question [1], press releases issued by the Ministry of Communications (DoT) regarding the blocking of 'spoofer calls' [25] [26] and RBI instructions [34] as well as the Supreme Court proceedings [28, 40, 42]. Case reports: News reports from similar digital arrest cases in Pune and Maharashtra to recreate the modus operandi [2] [9] [7] [8] [3] [32]. Research articles: journals articles that have passed the peer-review process covering social engineering, scam psychology, older people, telephony fraud and fraud detection (Section 2). Synthetic data: a 910-message corpus and a 22,605-transaction ledger yielded by the authors' code based on the patterns listed above (Section 4). None of the personal or banking information was used, it was just made-up.

3.3 Attack Reconstruction: The Digital Arrest Kill Chain

The attack was divided into six stages in accordance with the phases proposed in [12] and their idea of the kill-chain in [16] (shown in Fig. 2). Texts in Table II list what is generally reported to have occurred for each stage in documented cases of Pune; the persuasion principle used and the control that was not. The solution is based on this mapping as it indicates where a defender may be able to intervene.

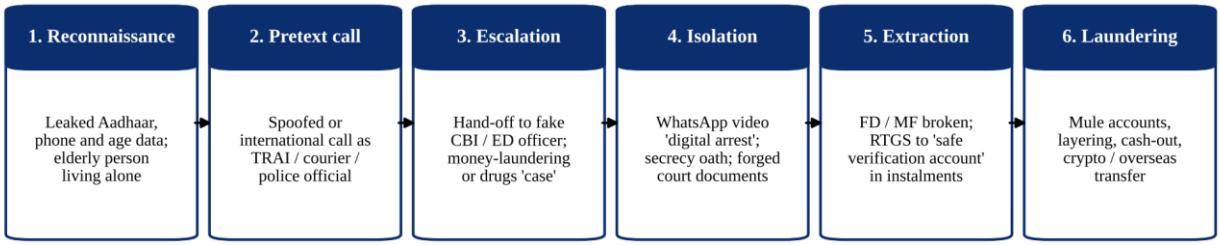


Fig. 2. Six-stage kill chain of the digital arrest scam, reconstructed from the problem statement and comparable documented cases.

Table II. Stage-by-stage reconstruction of the attack (details marked R are reconstructed from comparable cases)

Stage	What happens	Lever exploited	Control that failed
1. Reconnaissance	Victim profiled from leaked identity and phone data; an elderly person with savings is selected (R)	Information asymmetry	Data protection; leaked KYC data

Stage	What happens	Lever exploited	Control that failed
2. Pretext call	Phone call claiming to be from a courier, the telecom regulator or a police station; a package with drugs or misuse of Aadhaar is alleged [9], [3]	Authority, fear	Caller identity cannot be verified; spoofed or international numbers [25]
3. Escalation	Call handed to a fake senior officer of the CBI, ED or crime branch; a money-laundering case is described [7]	Authority, commitment	No independent channel to verify the 'officer'
4. Isolation	Victim moved to WhatsApp video; told she is under 'digital arrest', must keep the camera on and tell no one; forged warrants shown (R) [28]	Fear, secrecy, time pressure	Family not informed; platform cannot see call content
5. Extraction	Victim told to break deposits and transfer savings to a 'verification' account, with a promise of refund; several instalments over days (R) [2], [9]	Hope of refund, consistency	Bank processes out-of-pattern RTGS transfers to new payees
6. Laundering	Money moved through layers of mule accounts and withdrawn or converted [32]	-	Mule accounts opened with weak KYC; delayed reporting

From Table II two observations may be made. Stage 4 is the most important part of the attack for the victim; this is the attacker's most important asset. Second, the only time when a regulated institution with a duty of care is directly in the way of the attack and sees the victim's behaviour change is during Stage 5. This design puts the banking layer in a strong position, though, as a result.

3.4 Root-Cause Analysis

Five root causes were found. Voice and video channels are mostly unidentifiable and the first contact is made with a predetermined number that is spoofed or from abroad. (2) Deliberate isolation: when family members would have questions about the story, the literature identifies the strongest protection for older people would have been the demand of this secrecy, although they were deliberately removed [21]. (3) Speedy high value payments: RTGS and net banking are designed to deliver speed and execute withdrawal even if it is a wild one, once an authorised payment is made by the account holder. (4) Mule accounts: accounts can be stolen in a different name, and the account that a criminal receives may look ordinary [32] [8]. (5) Late reporting: victims are only able to report after the money has been layered, when recovery is difficult; the 1930 helpline reporting periods or "golden hour" rely on early reporting.

3.5 Proposed Solution: The DA-Shield Framework

DA-Shield (Digital Arrest Shield) is a five-layer structure that focuses on one or more layers of the kill chain (Fig. 3). The layers are supposed to overlap, in the event that the attack passes one layer it hits another.

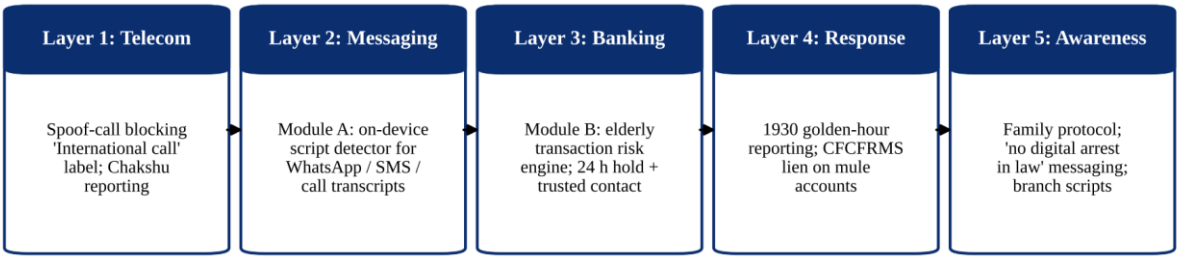


Fig. 3. The five layers of the proposed DA-Shield framework. Layers 2 and 3 are implemented in this study as Modules A and B.

Telecom (Stages 2-3): leverages an existing DoT spoof-call blocking and 'International Call' labelling [26] and suggests an on-device assistant to label calls from unknown numbers that include an arrest mention. Layer 2, Messaging (Stages 3-4) - Module A: an on-device classifier that warns the user (and family contact that has registered if agreed to by the user) when a digital-arrest script is received via WhatsApp and SMS text messaging (and, if agreed to by the user, via an audio phone transcript). The third module (Banking, Stage 5) – an explainable risk engine that is applied to all high value outgoing payments from an existing customer of age 60 and older. High scores mean that there is a 24 hour hold before the transaction is completed and a callback to a pre-registered trusted contact; medium scores means that there is a scripted conversation with a teller. Layer 4, Response (Stage 6): immediate filing through 1930 and the National Cyber Crime Reporting Portal to be able to freeze the receiving account while the money is still traceable [1]. Digitally arrested children are not in existence in the legal system and are upheld by a simple family protocol ('no agency arrests by video; call a family member before any transfer'), with a branch protocol and staff scripts, reinforcing official messaging that digital arrest does not exist in the law [3].

3.6 Module A Design: Scam-Script Detector

The detection component of Module A is a binary classification problem where the task is to classify a short message as either a digital-arrest script or not. Due to the absence of any public corpus of digital-arrets in the Indian context, a corpus was created with 13 Indian message families (as shown in Table III). Six patterns of scam families were identified from official advisories and news articles (courier drugs, Aadhaar money laundering, telecom disconnection, 'safe account' transfer, fake court order and threats to family). Then there are the seven legitimate families represented by alerts, family chats, courier updates, bank KYC reminders and services, and two families of hard negatives: police advisories, news items about digital arrest. The vocabulary used in hard negatives is the same as the scams ("CBI", "arrest", "digital arrest") and were intentionally added because a detector that constantly picks up every police warning would be of little practical use.

Table III. Message families in the synthetic corpus (70 messages each, 910 in total)

Family	Example (abridged)
S1 Courier/drugs	'A parcel booked on your Aadhaar to Taiwan contains MDMA... stay on video call, do not disconnect.'
S2 Aadhaar laundering	'Your Aadhaar is linked to a money laundering case of Rs 68 lakh. You are under digital arrest.'

Family	Example (abridged)
S3 Telecom	'Your mobile number will be disconnected in 2 hours as it is used for illegal activities.'
S4 Safe account	'Transfer your FD amount to RBI safe custody account. Money will be refunded within 48 hours.'
S5 Fake court	'Supreme Court order attached. Asset freeze under PMLA. Sign secrecy oath.'
S6 Family threat	'If you disconnect this call your son will also be arrested. Keep camera on.'
L1-L3, L5, L7 Legitimate	Bank debit alerts and OTPs; family chat (including Hinglish); courier delivery; KYC reminders; bills
L4 Police advisory (hard negative)	'There is no digital arrest in Indian law. Police never arrest on video call. Report fraud on 1930.'
L6 News (hard negative)	'Supreme Court orders CBI probe into digital arrest scams across country.'

Compared are three classifiers: (i) a baseline based on keyword-rule, (ii) TF-IDF word unigrams and bigrams in combination with Multinomial Naive Bayes, and (iii) the combination of TF-IDF word n-grams (n=1,2) and character n-grams (n=3,5) with class-balanced logistic regression. To deal with the spillover words and Hinglish words, Character n-gram were added to the model.

The evaluation protocol is the most critical design selection. A split of the whole train into an even number of messages would give scores of almost perfect recall of the same template messages and artificially inflate the scores. Instead, the families were split into seven folds, with one entire scam family and one entire legitimate family removed from each fold and the model trained on the other families in each fold. Each test message is thus sent from a script never previously encountered by the model, reflecting a new variant of the scam being seen in the wild. The pooled out-of-fold predictions were used for computing precision, recall, F1-score, accuracy and ROC-AUC. The primary metric used is recall - that is the percentage of scams that can be caught, since a missed scam can cost crores, a false alert is a moment of attention.

3.7 Module B Design: Elderly Transaction Risk Engine

The seven binary indicators, based on the reconstructed case and on documented cases, are used for scoring the outward transfer by a customer on Module B. The design of the engine needed to be made comprehensible: a branch officer needs to be able to explain to the customer why a payment is about to be held; and an auditor needs to be able to verify the decision. Every indicator has a weight and the score is the total of the weights of the indicators that fire (maximum is 100). The indicators and the reasons for the weighting of each indicator are summarized in Table IV.

Table IV. Risk indicators and weights used by Module B

Indicator	Weight	Rationale
Customer aged 60 or above	10	Senior citizens are the primary targets in Pune data [3]; low weight so age alone never triggers a hold

Indicator	Weight	Rationale
Amount at least 5 times the 90-day average outflow	20	Digital arrest instalments are far larger than normal spending [2]
Payee added recently (new beneficiary)	15	Victims are given fresh accounts for each instalment [9]
Payee account less than 180 days old	15	Mule accounts are often newly opened [32], [8]
Fixed deposit or mutual fund liquidated in last 48 hours	15	Victims are told to break deposits and sell investments [3]
Two or more high-value transfers in 72 hours	15	Scams extract money in repeated instalments
Staff observe customer on a call or being coached	10	Victims are often kept on the line while visiting the branch [28]

There are two thresholds that transform the score to an action. If a score is 40 or higher, a VERIFY action is triggered by which the teller reads a short message ("Has anyone on the phone asked you to make this payment?"). Ever get any instructions to keep it confidential (i.e., “Hasn't anyone told you to keep it secret?” The score of 60 or higher indicates a HOLD and the bank contacts a "trusted contact" that the customer has registered for the transfer to be delayed for 24 hours. The 24 Hour Hold is the basic intervention of this step, and it is important because it allows the victim to not interact with the scammer and allows a family member to be in the interaction instead which abolishes that isolation from Step 4. A machine learning comparison was created by training a random forest with the same features, with the split by customer so it wasn't captured twice in both the training and test sets.

3.8 Module C: Replay of the Reconstructed Case

As the actual instalment plan for the Pune case was not available, an indicative plan was drawn up: 6 RTGS transfers over 12 days (making a total of Rs. The amounts increase, then decrease and the total is 4.37 crore. New Payee accounts (weeks/month old), new deposits that are liquidated prior to most transfers (25, 48, 72, 95 and 112 lakh). This shape fits into the scenario of increasing instalments detailed in similar instances [2],[9]. The score of each transfer was computed by Module B to establish when DA-Shield would have stepped in to achieve the first point. This replay illustrates how the mechanism works, but is not intended to accurately set out the facts of the case.

4. Implementation Details

4.1 Environment and Tools

The prototype is executed in python 3 with scikit-learn [37] to extract features and models, NumPy for numerical computations and Matplotlib to create the figures. Scripts are divided into two: dashield_prototype.py (for generating the data, for Modules A, B and C, and for evaluation) and make_figures.py (for making figures). Each number in this report can be reproduced with a fixed seed (31) and a random seed.

4.2 Corpus Generation (Module A)

There are 3 to 5 message templates in each family, varying by agency name (CBI, ED, Mumbai Crime Branch, Narcotics Control Bureau, customs, TRAI), officer names, cities and amounts. The 70 messages per family were created by randomly filling in slots. A random message is deleted in 20% of cases and a random word is adjacent letter typoed in 15%, to prevent the classifier from analyzing and learning the clean formatting. To make it impossible to tell if a family-chat family or a scam family is just language, “kisi ko mat batana warna arrest hoga” (literal hesh = if you don't pay back a person, you'll be arrested) was added to both families and made into “Hinglish” templates.

4.3 Classifier Pipeline (Module A)

Algorithm 1: Module A training and leave-family-out evaluation

Input: Corpus of (text, label, family); Scam Families S1-S6; Legitimate Families L1-L7

1. Pick the scam family S(i mod 6) and legitimate family L(i mod 7) from the set of scam families and legitimate families, respectively, held out for fold i = 1 to 7.

3. Training set = all the messages that are not in the two held out families; test set = all the messages that are in these two held out families.

3. Joined TF-IDF word n-grams (of size (1,2)) and TF-IDF character n-grams (of size (3,5)) that use sublinear term frequency. 4. Fit logistic regression with C = 4, class_weight = balanced and max_iter = 2000; fit Naive Bayes with alpha = 0.3) on word TF-IDF.

6. Label and estimate the probability of labels for the test set; use the keyword-rule baseline (flag if 3 or more red-flag terms). Make predictions and append to pooled out-of-fold results.

7. Compute the precision, recall, F1 after all folds, accuracy, ROC-AUC and the confusion matrix for each model.

When deployed, the trained logistic regression model is small (a sparse weight vector) and, in turn, can be executed on a phone without any sending of the content of the messages to a server (important when it comes to privacy). If a message receives a score greater than 0.5 it displays a plain language warning ("Real police do not ask you to transfer money – call family or 1930. Do not accept messages of this nature.") and, if the user has added a family member to their account, a warning is sent to the family member but does not include any of the message content.

4.4 Risk Engine (Module B)

Algorithm 2: Module B scoring of one outward transfer

Calculate flag variable from the following information: Customer age, amount, length of average outflow in 90 days, flags indicating the payee, a flag indicating liquidation, counts of high value transfers within 72 hours, staff flag.

1. f_senior = (age >= 60); f_amount = (amount >= 5 x 90-day average)
2. f_newpayee = new Beneficiary; f_young = (payee account age < 180days) Last 48 h: 3. f_liquid = deposit or fund was liquidated; (high-value transfers in 72 h >= 2): 2. f_burst; staff observe coaching: 3. f_staff
4. score = 10 f_senior + 20 f_amount + 15 f_newpayee + 15 f_young + 15 f_liquid + 15 f_burst + 10 f_staff
5. If score equals or greater than 60: HOLD for 24 h, call trusted contact, explain to the customer the reasons. Otherwise, if score is >= 40: VERIFY using teller script. Else: ALLOW.
6. Record score, fire indicators and perform audit and model tuning action.

4.5 Transaction Simulation

The simulation included a block chain with 22,605 transactions.There were 22,605 transactions simulated in a block chain. The normal volume of activity was generated by 4000 customers, 60-90 year olds, with 3-8 transfers each. The monthly outflow had an average value that followed a log-normal distribution with a median of Rs. 30,000). All four per cent of legitimate transfers were actually large payments in a real test, such as hospital bills, education for a child or property, 45 per cent of which were to new payees and 30 per cent of which were a result of a deposit liquidation; in all, eight per cent of legitimate payees had new accounts and 2 per cent of legitimate customers were identified as being on a call by staff. There were 150 digital-arrest campaigns (2 to 9 instalments of Rs. 3 lakh to Rs. 1.2 crore. Preventing fraud being easily identified, 85% of its instalments were paid to new payees, 70% to ‘young' accounts, 65% after a liquidation and 25% were identified by staff. There were 793 fraudulent transactions (3.5%).

4.6 Deployment Architecture and Privacy

In production, Module B is hosted within the core banking system or the payment gateway fraud engine of the bank, along with the pre-existing early-warning systems mandated by RBI [34]. It would be gathered at the time of opening the account or at the time of regeneration/renewal of the customer before application of the hold on the amount to be transferred overseas, hence the routine payment like bills and pension payments is not affected. Module A would execute on the user's device, and no data about the contents of the messages would be uploaded, only a yes/no alert would be shared with the registered contact. With the purpose-limitation principle of India's data protection law [44] in focus, this design limits personal data processing.

4.7 Reproducibility

Running “python3 dashield_prototype.py” will re-create the corpus, re-create the ledger and all of the metrics in Section 5, then output them to results.json. The code "python3 make_figures.py" can be used to recreate the Figs. 1 to 6. The code will be provided upon request by the group.

5. Results and Discussion

5.1 Module A: Scam-Script Detection

The pooled Leave Family Out results over 980 test predictions (490 scam and 490 legitimate) are reported in Table V and Fig. 4 (the courier family goes in two folds).

Table V. Module A performance under leave-family-out cross-validation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Keyword-rule baseline	0.710	0.789	0.573	0.664	-
TF-IDF + Naive Bayes	0.800	0.715	0.998	0.833	0.872
TF-IDF + Logistic Regression	0.819	0.751	0.955	0.841	0.840

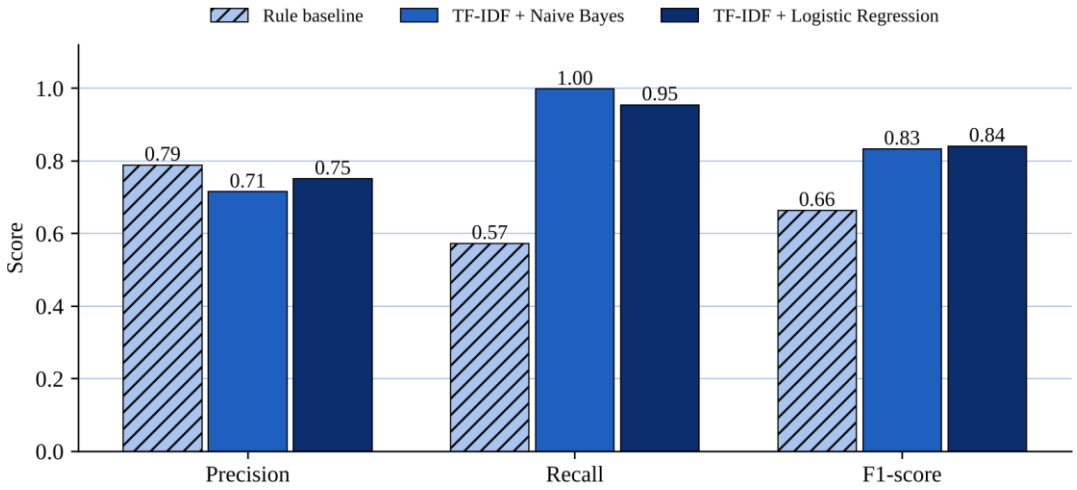


Fig. 4. Precision, recall and F1-score of the three Module A detectors on unseen message families.

The logistic regression model achieved the best balance, while recalling 468 of 490 scam messages containing scripts that it has never encountered, it missed 22. Naive Bayes had the highest recall of 0.998, and raised 195 false alarms compared to 155 false alarms for logistic regression. The best-precision model was baseline (0.789) with the lowest recall (0.573) of the 209 scam messages. The words which are less apparent like “arrest” or “CBI” are the ones that are missed by its failures, concentrated in the ‘family threat', ‘safe account' and ‘fake court' families. This is an important discovery: There are scripts that do not mention the word secret but they still are dangerous and a set list of keywords does not detect them. The error analysis also demonstrates that the place at which the learned models have a problem! About 72% of the logistic regression model's false positives (111 of 155) were police advisories and news reports, the two hard-negative families. If a model hasn't received a genuine warning from either the CBI or the ED saying to stop transferring money over phone, the fact that a warning was sent does closely resemble a scam. The performance on the folds ranged between a minimum of 0.66 for the fold that contained the 'safe account' scams with the police advisories, and a maximum of >0.94 for folds containing the 'courier/truck' or ‘telecom' scripts. In practice, it is simple: there are only a few verified senders (police, banks, I4C) so both the classifier and sender verification can be used together; the true advisories should be used in training. Recalls, as a form of behaviour, are OK if you are providing a warning, because if you do not warn them and they fall for the scam, the potential for loss is much greater.

5.2 Module B: Transaction Risk Engine

Table VI. Module B performance on 22,605 simulated transactions (793 fraudulent)

Configuration	Precision	Recall	F1	False-positive rate
Rule engine, VERIFY (score >= 40)	0.491	0.999	0.659	3.76%
Rule engine, HOLD (score >= 60)	0.763	0.950	0.846	1.07%
Rule engine, HOLD (customer-held-out test set)	0.819	0.960	0.884	0.88%
Random forest (customer-held-out test set)	0.908	0.997	0.950	0.42%

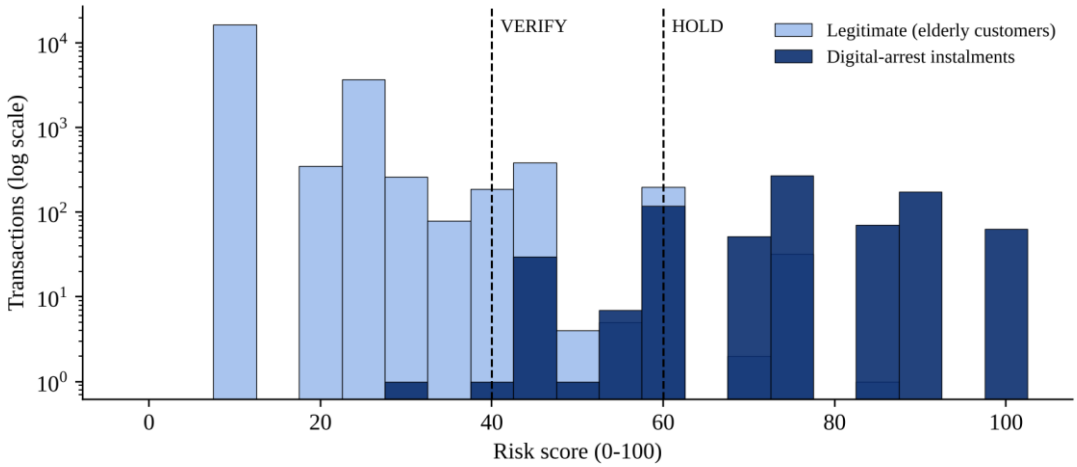


Fig. 5. Distribution of Module B risk scores for legitimate and fraudulent transfers (log scale), with the VERIFY and HOLD thresholds.

The rule engine retained 95.0% of the fraudulent instalments and 1.07% of the legitimate transactions at the HOLD threshold and obtained a score of 0.998 at the ROC-AUC measure, distinguishing the two classes (Fig. 5). This is more powerful than when looking at each individual transfer: 122 of the 150 simulated campaigns (81.3%) were performed at the first instalment, and all campaigns were performed at an instalment. If every hold is successful in busting the scam, Rs. 484.44 crore of the Rs. The targeted 502.88 crore would have been left in the victims' accounts (96.3%). Campaigns which made it past the first check were generally sent to an older payee account without the deposit liquidation; their second instalment would then be trapped by the 72 hour burst indicator, hence the need for repeated-transfers monitoring. Our random forest classifier had a higher F1 score than the rule engine on the held-out customers (0.950 vs 0.884) and fewer false positives (1/2). This result should be taken with a pinch of salt: the indicators used for the simulation are the very same used by the models, which makes the classes well-separated and explains why the forest's near-perfect ROC-AUC (1.000). The authors thus suggest that the transparent rule engine be deployed first, as all the holds can be communicated to the customer, and audited, plus a learned model be deployed afterwards as a second opinion once real, labelled outcomes are present. The effect of false-positive results was also studied. The total number of legitimate transfers that would be delayed by up to 24 hours during the simulation period is approximately 233 transfers from senior citizens. Most of these are actually unusual payments that quite possibly are huge ones, and banks tend to examine these payments. An inconvenience to getting paid by several lakh rupees a few days later is an inconvenience, but losing the money for a lifetime is not

an inconvenience. The trusted-contact callback also allows for a legitimate hold to be released in mere minutes.

5.3 Module C: Replay of the Pune Case

Table VII. Module B decision for each instalment of the reconstructed Pune case (illustrative schedule)

Step	Day	Amount (Rs. lakh)	Cumulative (Rs. lakh)	Score	Decision
1	1	25	25	75	HOLD 24h + trusted-contact callback
2	2	48	73	90	HOLD 24h + trusted-contact callback
3	4	72	145	90	HOLD 24h + trusted-contact callback
4	6	95	240	75	HOLD 24h + trusted-contact callback
5	9	112	352	90	HOLD 24h + trusted-contact callback
6	12	85	437	75	HOLD 24h + trusted-contact callback

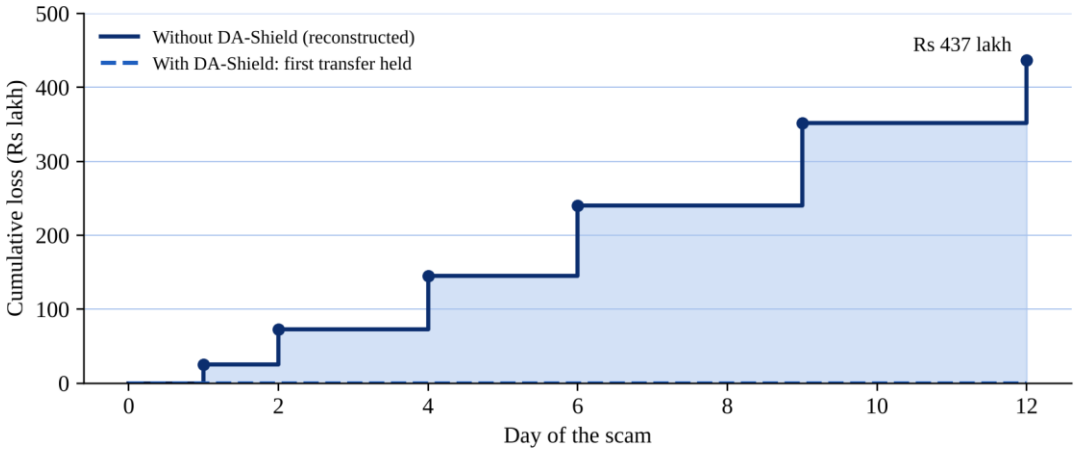


Fig. 6. Cumulative loss in the reconstructed case, without DA-Shield and with the first transfer held.

All the entries in the revised program were assigned 75 or 90, above the HOLD level (Table VII). The very first transfer of Rs. The 25 lakh caused five red flags: the age of the customer, an amount much lower than the customer's regular outflow, a new payee, a payee that's no more than a few weeks old and a deposit that broke the previous day. DA-Shield would have the transaction pending for 24 hours and a family member would have been notified. The scam relies on the victim speaking to no one; one phone call from the bank to a son or daughter is likely to terminate the scam. Even in the worst case, if the first transfer was sent even before the hold the second transfer would have been also held, thus the

loss would have been Rs. The highly valued 25 lakh and the protection of Rs. 4.12 crore (about 94%) of the Rs. 4.37 crore (Fig. 6).

5.4 Discussion

The results lead to three conclusions. The first thing is that the bank is the best control point. Although the telecoms layer is already preventing the majority of spoofed calls [26], losses are increasing year on year [1] as fraudsters continue to seek out other methods to send calls, such as using WhatsApp and international numbers. Scammers can register new accounts on messaging platforms [42] while accounts can be banned. But the victim's bank is aware of the one thing the scammer can't at all hide – that a senior citizen suddenly moved some of his savings into some unknown bank account. Second – The intervention should help to overcome the isolation, not simply add a warning. If the frightened person can back out of the area, a fake officer can monitor them on video, but a delay on the situation, plus a call to a family member, changes the situation. Third, the ability to detect text scams should be evaluated on realistic look-alikes. The LFO results reveal that while some evaluations might seem straightforward, the reality is that it is more difficult to know if a warning is real than if it is a scam.

The findings corroborate other literature. DA-Shield is based on the ideas of building systems around predictable human behaviour, as put forward by Stajano and Wilson [17] and social support as protective as pointed out by James et al. [21]. The judiciary's interest in bank accounts that are targeted in such scams [40] is another reason why banks should be an active line of defense in both the receiving direction (mule detection [33]) and as suggested here in the sending direction.

5.5 Recommendations

Table VIII. Recommendations for each stakeholder

Stakeholder	Recommendation
Banks	Use a senior-citizen transfer-risk check, with a 24-hour interval for the transfer, and have a trusted-contact callback for high-risk transfers; educate branch staff on how to identify a person who is being coached on the telephone.
RBI	Make provision for a voluntary 'cooling-off' facility for high value transfers for senior citizens and share mule-detection with sending banks.
Telecom operators and DoT	Continued with spoof blocking; Internationally made calls: Give a clear warning about arrest/police; Chakshu reporting promoted.
Messaging platforms	Provide on-device scam-script detection and 'trusted contact' for the older user; identify accounts impersonating agencies.
Police and I4C	Continue emphasizing need to post case alerts promptly; enhance 1930 golden hour response.
Families	Make a family agreement – no transfer of savings funds without a call from a family member; sign up as a trusted contact with the bank.

5.6 Limitations

- A detailed public record of this particular attack was not found, so the attack timeline and instalment schedule is reconstructed and illustrative. The messages are all synthetic as well as the transaction ledger. They are constructed from documented patterns and contain intentionally challenging instances, but real data will be messier and the instances will not produce the positive metrics reported, particularly for a random forest. Module A is Analysis of Texts only. Much of

a digital arrest occurs on voice and video calls and future work would involve introducing the detector to speech-to-text transcriptions, with user consent. The rule weights were determined using domain reasonings, not disease data, and need tuning on real bank data. A hold could be circumvented by having the victim use multiple banks or get to a bank's location themselves (being coached to do so); sharing of risk signals in the inter-bank sector would be needed to fill the gap.

6. Conclusion

The example we were given was the case of the Pune digital arrest scam from January of last year; for which the old woman lost Rs. 4.37 crore, represents a fraud without malware, without any software vulnerability. This approach involves pretending to be an authority, confining the victim and using the fast pace of India's payment system against her. According to national statistics, such fraud has increased from Rs. 91 crore in reported losses in 2022 to more than Rs. In 2024, the amount is 1,935 crore [1] and the proportion of senior citizens in Pune is disproportionately high [3]. This report re-constructed the attack's kill chain into six stages, and determined the five root causes of this incident, providing a proposal of DA-Shield, a 5-layer defence. There were two components implemented and tested. A TF-IDF and logistic regression detector (Module A) made an F1-score of 0.84 and a recall of 0.95 on unseen scam scripts, outperforming a baseline with only keywords that made an F1-score of 0.66 and a recall of 0.88; it was also found that the majority of false alarms were generated by genuine police warnings. The explainable risk engine for transfers for senior citizens achieved excellent results, capturing 95.0% of fraudulent instalments and preventing 81% of simulated frauds at the first instalment with 1.07% false positive rate. Replayed on the reconstructed Pune case, it would have transferred the first case and safeguarded at least 94% of the money. The main point is that if someone was being led by a bogus officer she can't be expected to realise that she is being lead, the system around her has to create a pause and introduce a person who is trusted by her. There is a need to further develop Module A for transcribed voice calls and regional languages, test Module B on anonymised real transaction data in collaboration with a bank, and investigate the possibility of inter-bank sharing of risk signals, which will make it impossible to circumvent a bank's hold at another bank.

References

[1]Ministry of Home Affairs, Government of India, "Combating digital arrest, blackmail, and cyber impersonation of law enforcement," Rajya Sabha Unstarred Question No. 1505, answered 12 Mar. 2025. [Online]. Available: https://rsdebate.nic.in/bitstream/123456789/757210/1/PQ_267_12032025_U1505_p268_p273.pdf

[2]A. Nikam, "Woman duped of Rs. 4.82 crore in 'digital arrest' cyber fraud in Pune," The Bridge Chronicle, 21 Jan. 2026. [Online]. Available: https://www.thebridgechronicle.com/pune/woman-duped-digital-arrest-cyber-fraud-pune-agn97

[3]Punekar News, "Senior citizens lose Rs 100 crore in 136 digital arrest scams in Pune in 19 months," 21 Sep. 2026. [Online]. Available: https://www.punekarnews.in/?p=245348

[4]Press Trust of India, "Digital arrests, cyber crimes tripled during 2022-24, defrauded amount jumped 21 times: Govt," The Week, 12 Mar. 2025. [Online]. Available: https://www.theweek.in/wire-updates/national/2025/03/12/del54-rsq-cybercrime.html

[5]Inc42, "Over 7 lakh SIMs, 83K WhatsApp accounts blocked till February: Govt," Inc42 Media, Mar. 2025. [Online]. Available: https://inc42.com/?p=506944

[6]The420.in, "Pune senior citizens lose nearly Rs. 100 crore in 136 digital arrest scams over 19 months," Sep. 2026. [Online]. Available: https://the420.in/?p=95567

[7]M. Ansari, "Pune Cyber Police bust 'digital arrest' scam; two arrested for defrauding senior citizen of Rs 10.74 crore," Punekar News, 28 Feb. 2026. [Online]. Available: https://www.punekarnews.in/?p=229619

[8]Press Trust of India, "Social media influencer held in Rs 10.7 crore digital arrest scam," ThePrint, 13 Aug. 2026. [Online]. Available: https://theprint.in/india/social-media-influencer-held-in-rs-10-7-crore-digital-arrest-scam/3014171/

[9]Press Trust of India, "Octogenarian dies weeks after losing Rs 1.2 crore in 'digital arrest' fraud," The Tribune. [Online]. Available: https://www.tribuneindia.com/news/india/octogenarian-dies-weeks-after-losing-rs-1-2-crore-in-digital-arrest-fraud (accessed Oct. 14, 2026).

[10] R. B. Cialdini, Influence: Science and Practice, 5th ed. Boston, MA, USA: Pearson, 2009.

[11] C. Hadnagy, Social Engineering: The Science of Human Hacking, 2nd ed. Indianapolis, IN, USA: Wiley, 2018.

[12] F. Mouton, L. Leenen, and H. S. Venter, "Social engineering attack examples, templates and scenarios," Computers & Security, vol. 59, pp. 186-209, 2016.

[13]K. Krombholz, H. Hobel, M. Huber, and E. Weippl, "Advanced social engineering attacks," Journal of Information Security and Applications, vol. 22, pp. 113-122, 2015.

[14]F. Salahdine and N. Kaabouch, "Social engineering attacks: A survey," Future Internet, vol. 11, no. 4, Art. no. 89, 2019.

[15] M. Workman, "Wisecrackers: A theory-grounded investigation of phishing and pretext social engineering threats to information security," Journal of the American Society for Information Science and Technology, vol. 59, no. 4, pp. 662-674, 2008.

[16] E. M. Hutchins, M. J. Cloppert, and R. M. Amin, "Intelligence-driven computer network defense informed by analysis of adversary campaigns and intrusion kill chains," Leading Issues in Information Warfare & Security Research, vol. 1, no. 1, pp. 80-106, 2011.

[17] F. Stajano and P. Wilson, "Understanding scam victims: Seven principles for systems security," Communications of the ACM, vol. 54, no. 3, pp. 70-75, 2011.

[18] M. T. Whitty, "The scammers persuasive techniques model: Development of a stage model to explain the online dating romance scam," British Journal of Criminology, vol. 53, no. 4, pp. 665-684, 2013.

[19] M. Button, C. Lewis, and J. Tapley, "Fraud typologies and the victims of fraud: Literature review," National Fraud Authority, London, U.K., 2009.

[20] D. Burnes, C. R. Henderson Jr., C. Sheppard, R. Zhao, K. Pillemer, and M. S. Lachs, "Prevalence of financial fraud and scams among older adults in the United States: A systematic review and meta-analysis," American Journal of Public Health, vol. 107, no. 8, pp. e13-e21, 2017.

[21] B. D. James, P. A. Boyle, and D. A. Bennett, "Correlates of susceptibility to scams in older adults without dementia," Journal of Elder Abuse & Neglect, vol. 26, no. 2, pp. 107-122, 2014.

[22]R. N. Spreng, J. Karlawish, and D. C. Marson, "Cognitive, social, and neural determinants of diminished decision-making and financial exploitation risk in aging and

dementia: A review and new model," Journal of Elder Abuse & Neglect, vol. 28, no. 4-5, pp. 320-344, 2016.

[23] N. C. Ebner et al., "Uncovering susceptibility risk to online deception in aging," The Journals of Gerontology: Series B, vol. 75, no. 3, pp. 522-533, 2020.

[24]H. Tu, A. Doupe, Z. Zhao, and G.-J. Ahn, "SoK: Everyone hates robocalls: A survey of techniques against telephone spam," in Proc. IEEE Symposium on Security and Privacy, San Jose, CA, USA, 2016, pp. 320-338.

[25] Department of Telecommunications, Ministry of Communications, "Government issues directions for blocking of incoming international spoofed calls displaying Indian mobile numbers," Press Information Bureau, New Delhi, 26 May 2024.

[26]Department of Telecommunications, "Department of Telecom (DoT) acts pro-actively to protect citizens from cyber frauds through spoofed calls," Press release, Dec. 2024. [Online]. Available: https://sancharsaathi.gov.in

[27] M. Westerlund, "The emergence of deepfake technology: A review," Technology Innovation Management Review, vol. 9, no. 11, pp. 39-52, 2019.

[28]R. Choudhury, "Supreme Court takes suo motu cognisance of digital arrest scams, seeks response from Centre, CBI," Bar and Bench / Moneylife, 17 Oct. 2025. [Online]. Available: https://moneylife.in/article/supreme-court-takes-suo-motu-cognisance-of-digital-arrest-scams-seeks-response-from-centre-cbi/78615.html

[29]A. Abdallah, M. A. Maarof, and A. Zainal, "Fraud detection system: A survey," Journal of Network and Computer Applications, vol. 68, pp. 90-113, 2016.

[30]A. Dal Pozzolo, O. Caelen, Y.-A. Le Borgne, S. Waterschoot, and G. Bontempi, "Learned lessons in credit card fraud detection from a practitioner perspective," Expert Systems with Applications, vol. 41, no. 10, pp. 4915-4928, 2014.

[31]F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in Proc. 8th IEEE International Conference on Data Mining (ICDM), Pisa, Italy, 2008, pp. 413-422.

[32]All India Radio News, "Maharashtra Police arrest seven in Rs 58 crore digital arrest scam," 17 Oct. 2025. [Online]. Available: https://www.newsonair.gov.in

[33]Business Standard, "Explained: RBI is using an AI tool MuleHunter.ai to cut down digital frauds," Dec. 2024. [Online]. Available: https://www.business-standard.com/finance/personal-finance/explained-rbi-has-a-new-ai-tool-mulehunter-ai-to-reduce-digital-frauds-124120900250_1.html

[34] Reserve Bank of India, "Master Directions on Fraud Risk Management in Commercial Banks (including Regional Rural Banks) and All India Financial Institutions," RBI/DOS/2024-25/118, Mumbai, 15 Jul. 2024.

[35]T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, "Contributions to the study of SMS spam filtering: New collection and results," in Proc. 11th ACM Symposium on Document Engineering (DocEng), Mountain View, CA, USA, 2011, pp. 259-262.

[36] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," Information Processing & Management, vol. 24, no. 5, pp. 513-523, 1988.

[37] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[38] Government of India, The Information Technology Act, 2000 (Act No. 21 of 2000), as amended by the Information Technology (Amendment) Act, 2008.

[39] Government of India, The Bharatiya Nyaya Sanhita, 2023 (Act No. 45 of 2023).

[40]Onmanorama, "Supreme Court orders nationwide CBI probe into digital arrest scams," 1 Dec. 2025. [Online]. Available: https://www.onmanorama.com/news/india/2025/12/01/digital-arrest-scams-cbi-investigation-sc-order.html

[41]Storyboard18, "Supreme Court grants CBI full authority to tackle nationwide 'digital arrest' scams," 2 Dec. 2025. [Online]. Available: https://www.storyboard18.com/digital/supreme-court-grants-cbi-full-authority-to-tackle-nationwide-digital-arrest-scams-85176.htm

[42]Moneylife Digital Team, "Digital arrests: Supreme Court calls for separate offence, harsher punishment against accused," Moneylife, 28 Jul. 2026. [Online]. Available: https://www.moneylife.in/article/digital-arrests-supreme-court-calls-for-separate-offence-harsher-punishment-against-accused/81184.html

[43] Federal Bureau of Investigation, "Internet Crime Report 2024," Internet Crime Complaint Center (IC3), Washington, DC, USA, 2025.

[44] Government of India, The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023).